

Week 7 In-Class Worksheet

Nonparametric Tests

ANSWER KEY

Course: H524 Introduction to Biostatistics

Topic: Sign Test, Wilcoxon Tests, Kruskal-Wallis, Normality Assessment

1 Problem 1: Sign Test

Recovery times: 12, 15, 10, 13, 16, 11, 14, 9, 12, 15

Part A: Hypotheses

H_0 : Median recovery time = 14 days

H_A : Median recovery time $\neq$ 14 days (one-sided)

Part B: Calculate signs

Differences from 14: -2, 1, -4, -1, 2, -3, 0, -5, -2, 1

After removing zero: -2, 1, -4, -1, 2, -3, -5, -2, 1

$B^+ = 3$ (positive signs), $B^- = 6$ (negative signs), $B = \min(3, 6) = 3$

Part C: R code and p-value

```
recovery <- c(12, 15, 10, 13, 16, 11, 14, 9, 12, 15)
diff <- recovery - 14
signs <- sign(diff[diff != 0])
b_minus <- sum(signs < 0) # Count negative (below 14)
n <- length(signs)

# One-sided test: testing if median < 14
# We expect MORE negative signs if median < 14
binom.test(b_minus, n, p=0.5, alternative="greater")
```

P-value: 0.2539

Part D: Conclusion

Since $p = 0.2539 > 0.05$, we fail to reject H_0 . There is no evidence that the median recovery time is less than 14 days.

2 Problem 2: Wilcoxon Signed-Rank Test

Part A: Differences and ranks

Differences: $-2, 1, -4, -1, 2, -3, 0, -5, -2, 1$ (excluding zero: $-2, 1, -4, -1, 2, -3, -5, -2, 1$)

Difference	Diff	Rank	Signed Rank
-2	2	5	-5
1	1	2	2
-4	4	8	-8
-1	1	2	-2
2	2	5	5
-3	3	7	-7
-5	5	9	-9
-2	2	5	-5
1	1	2	2

Part B: Sum of ranks

$$W^+ = 2 + 2 + 5 = 9$$

$$W^- = |-5 - 8 - 2 - 7 - 9 - 5 - 2| = 36$$

Part C: R code and p-value

```
wilcox.test(recovery, mu=14, alternative="two.sided")
```

P-value: 0.1209

Part D: Comparison

The Wilcoxon test is more powerful because it uses both direction AND magnitude of differences. The sign test only uses direction, while Wilcoxon is more sensitive to the pattern in the data.

3 Problem 3: Paired Data

Part A: Differences

Participant	1	2	3	4	5	6	7	8
Difference	4	-2	7	3	5	3	2	4

Part B: Sign test

```
bp_before <- c(142, 138, 155, 148, 160, 145, 152, 149)
bp_after  <- c(138, 140, 148, 145, 155, 142, 150, 145)
bp_diff   <- bp_before - bp_after
signs     <- sign(bp_diff[bp_diff != 0])
b_plus    <- sum(signs > 0)
binom.test(b_plus, length(signs), p=0.5,
            alternative="two.sided")
```

$B^+ = 7$, $B^- = 1$, **p-value: 0.0703**

Part C: Wilcoxon signed-rank test

```
wilcox.test(bp_before, bp_after, paired=TRUE)
```

P-value: 0.0245

Part D: Recommendation

The Wilcoxon signed-rank test is recommended. It uses the magnitude of differences (which vary from 2 to 7 mmHg), making it more powerful. Wilcoxon $p = 0.0245$ (significant at $\alpha = 0.05$), while sign test $p = 0.0703$ (not significant).

4 Problem 4: Two-Sample Wilcoxon Rank-Sum Test

Part A: R code

```
med_A <- c(6, 7, 5, 8, 6, 7, 9)
med_B <- c(4, 5, 6, 5, 7, 4, 6)
wilcox.test(med_A, med_B, alternative="two.sided")
```

W statistic: 40 P-value: 0.0502

Part B: Conclusion

Since $p = 0.0502 > 0.05$, we fail to reject H_0 at the 0.05 level. No significant difference between medications (though very close to significant).

Part C: Why use this test

- Pain scores are ordinal (0-10 scale), not truly continuous
- Small sample sizes ($n=7$ each) make normality hard to assess
- Nonparametric test doesn't assume normal distributions

5 Problem 5: Kruskal-Wallis Test

Part A: R code

```
low_dose <- c(5.2, 6.1, 5.8, 6.3, 5.5)
med_dose <- c(6.8, 7.2, 7.5, 6.9, 7.1)
high_dose <- c(7.8, 8.2, 8.5, 7.9, 8.1)

sleep_hours <- c(low_dose, med_dose, high_dose)
dose <- factor(rep(c("Low", "Medium", "High"), each=5))

kruskal.test(sleep_hours ~ dose)
```

H statistic: 12.50 P-value: 0.0019

Part B: Conclusion

Since $p = 0.0019 < 0.05$, we reject H_0 . Strong evidence that at least one dosage level differs in effectiveness for sleep duration.

Part C: Follow-up test

Perform pairwise Wilcoxon rank-sum tests with Bonferroni correction:

```
pairwise.wilcox.test(sleep_hours, dose,
                     p.adjust.method="bonferroni")
```

6 Problem 6: Checking Normality Assumptions

Part A: Shapiro-Wilk test

```
high_dose <- c(7.8, 8.2, 8.5, 7.9, 8.1)
shapiro.test(high_dose)
```

W statistic: 0.9636 P-value: 0.8327

Part B: Can we assume normality?

Yes, $p = 0.8327 > 0.05$, so we fail to reject the null hypothesis of normality. The data are consistent with a normal distribution.

Part C: Q-Q plot

```
qqnorm(high_dose)
qqline(high_dose, col="red")
```

Points should fall close to the red reference line, supporting the Shapiro-Wilk result that the data appear normally distributed.

Part D: Was Kruskal-Wallis appropriate?

Yes, Kruskal-Wallis was still appropriate because:

- With only $n=5$ per group, normality is hard to assess reliably
- We would need to check normality for ALL three groups
- Kruskal-Wallis is conservative - minimal power loss if data are normal

7 Problem 7: Choosing the Right Test

Part A: Survival times (highly skewed), two groups

Answer: Wilcoxon rank-sum test (Mann-Whitney U test)

Reason: Two independent groups, highly skewed data violates normality assumption for t-test.

Part B: Median readmissions vs. 3, one sample

Answer: Sign test or Wilcoxon signed-rank test

Reason: One-sample test for median. Wilcoxon more powerful if magnitude matters.

Part C: Average weight loss, four diets, normal data

Answer: One-way ANOVA (parametric)

Reason: Four groups, data appear normal. ANOVA more powerful when assumptions met.

Part D: Pre/post depression scores, ordinal scale

Answer: Wilcoxon signed-rank test

Reason: Paired data, ordinal scale (1-10 rating) makes ranks appropriate.